

HARI KRISHNA SHAH

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Professional Summary

Full Stack Software Engineer with expertise in **MERN stack, system design, and scalable architectures**. Experienced in building enterprise-grade applications serving **25+ organizations**. Strong foundation in **data structures, algorithms**, and **full-stack development** with proven track record of delivering high-impact solutions.

Education

National Institute of Technology, Rourkela

M.Tech in Computer Science and Engineering

Vellore Institute of Technology

B.Sc. Computer Science — GPA: 9.72/10 — **Gold Medalist**

Aug 2025 – Aug 2027

Rourkela, Odisha

Aug 2021 – May 2024

Vellore, Tamil Nadu

Professional Experience

Diginirman

June 2024 – Present

Lead Full Stack Developer

Kathmandu, Nepal

- Architected **Palika GIS** (palikagis.com), serving **25+ municipalities**, generating **2+ crore revenue in first year**.
- Designed and developed **Diginirmaan Official Website** (diginirmaan.com.np), establishing company's digital presence and showcasing product portfolio.
- Built **Emergency Management System** with real-time incident reporting (fire, road damage, etc.) via Socket.io; displays live GPS coordinates, severity, images/videos at management office for immediate action.
- Developed **House Numbering System** generating metric number plates based on proximity algorithms from main/connecting roads; visualizes house locations with assigned numbers on GIS interface.
- Engineered **Road Visualization System** mapping municipality roads with metadata (type, length, width, drainage, pavement type); integrated with GIS layers for infrastructure planning.
- Implemented **database caching** for computed stats, **spatial filters** using PostGIS; built **Digital Survey** with live logging and batch processing for **100K+ responses**.
- Created **WebSocket download system**, **auto Dropbox backup** with status monitoring, and **collaborative land-zone recommendation** with multi-party live commenting.

Vbay

May 2024 – June 2024

Co-Founder

Kathmandu, Nepal

- Built and launched Vbay during college, gaining end-to-end product experience across **RESTful APIs, UI/UX, and performance optimization**.
- Evaluated both engineering and business outcomes, developing a holistic understanding of product development before shutting down due to college regulations.

Key Technical Projects

Palika GIS | *React, Node.js, PostgreSQL, PostGIS, Socket.io, Leaflet.js*

(View: www.palikagis.com)

- **Emergency Management:** Real-time incident reporting system with Socket.io; residents submit emergencies (fire, damage) with GPS, severity, media; live dashboard for management office enables immediate response and incident resolution.
- **House Numbering:** Algorithmic metric number plate generation based on proximity from main/connecting roads; GIS visualization of houses with assigned numbers for municipal addressing system.
- **Road Visualization:** Complete municipality road network mapping with attributes (type, length, width, drainage, pavement); integrated layers for infrastructure planning and maintenance tracking.
- **Digital Profile:** Advanced filtering by 15+ demographics (caste, language, disease, vehicle); generates ward/tole reports.
- **Digital Survey:** Live logging, batch processing for **100K+ responses**, WebSocket download system, auto Dropbox backup with status monitoring.

Tetra Labs | *Next.js, TypeScript, PostgreSQL, GraphQL*

(View: tetralabs.diginirmaan.com.np)

- Civil testing lab SaaS enabling clients/contractors to request tests (compressive, tensile, base strength); automated workflow from test request → lab data entry → report generation → stakeholder download portal with RBAC.

Diginirmaan Official Website | *React, Next.js, Tailwind CSS*

(View: www.diginirmaan.com.np)

- Company website showcasing product portfolio, services, and client testimonials; responsive design with optimized performance.

Municipal Platform | *React, Node.js, Express, MongoDB*

(View: lalitpur.palikagis.com)

- Modular backend for content management, events, employee directory; adopted by multiple municipalities.

Technical Skills

Languages: TypeScript, JavaScript, Python, C, C++, SQL
Frontend: React.js, Next.js, Redux, Tailwind CSS, Bootstrap, Chakra UI
Backend: Node.js, Express.js, GraphQL, TypeGraphQL, RESTful APIs, Socket.io, WebSockets
Databases: PostgreSQL, PostGIS, MongoDB, MySQL, TypeORM
Tools: Git, Docker, AWS, DigitalOcean, Leaflet.js, Linux, Jira, Dropbox API
Core CS: Data Structures, Algorithms, System Design, DBMS, Computer Networks

Certifications

MERN Stack Development – 135 hours, Broadway Infosys	Verify
Front-End Web Development – 90 hours, Broadway Infosys	Verify
C & C++ Training – Broadway Infosys	Verify